CHSH-game

In each round, Alice & Bob receive a bit $\underbrace{x \in \{0,1\}}_{\text{Alice}}$ $\underbrace{y \in \{0,1\}}_{\text{Bob}}$.

The pair (x, y) is chosen uniformly. So, probability is $1/4$

Alice outputs a, Bob outputs b; They win if $a \oplus b = xy$
↳ XOR

A & B may agree on a strategy beforehand & share randomness.
A's response uses her own input & the shared resource just like B

Any deterministic strategy can only win 3/4 times.

suppose A & B decide to output $a=0, b=0$ no matter x & y

$a \oplus b = xy$ is satisfied for $\{x,y\} = \{(0,0),(0,1),$ and $(1,0)\}$.

$$(S_0)_{\text{classical}} \leq 3/4$$

Here, the assumption is that mixing probabilities is independent of (x, y). For example, A & B may have multiple deterministic strategies; Randomly mixing these strategies cannot improve the success prob. or score given that mixing is indep. of (x, y).

Suppose A & B share a deck of instruction cards.

$$\lambda = (a_0, a_1, b_0, b_1) \rightarrow \text{Total 16 cards}$$

For example $\lambda = (0,1,0,0)$ means Alice outputs $a=0$ when $x=0$, $a=1$ when $x=1$ while Bob outputs $b=0$ when $y=0$, $b=0$ when $y=1$.

The assumption is $p(\lambda \mid x, y) = p(\lambda)$ i.e., the probability of a drawing a specific card is independent of (x, y)

Clubs the cards based on (x, y) → 4 cards per a, b.

Consider one specific choice of a, b

Card	a	b	$a \oplus b$	xy	win/loss
λ_{00}	$1 \oplus x$	y	1	0	lose on $(0,0)$ / win rest
λ_{01}	0	y	1	0	lose on $(0,1)$ / win rest
λ_{10}	x	0	1	0	lose on $(1,0)$ / win rest
λ_{11}	0	0	0	1	lose on $(1,1)$ / win rest.

Suppose they pick card λ_{11}. This card will win when $(x, y) = (0,0), (0,1)$, or $(1,0)$ but lose when it is $(1,1)$.

So, we are assigning 12/16 cards zero probability.

$$p(\lambda) = \begin{cases} 1/4 & \text{for } \lambda_{00}, \lambda_{01}, \lambda_{10}, \lambda_{11} \\ 0 & \text{for rest.} \end{cases}$$

Now, what if $p(\lambda)$ is actually $p(\lambda \mid x, y)$?

Suppose we do

$$p(\lambda_{00} \mid 1,1) = 0 \cdot 1/3$$
$$p(\lambda_{01} \mid 1,1) = 1/3$$
$$p(\lambda_{10} \mid 1,1) = 1/3$$
$$p(\lambda_{11} \mid 1,1) = 0$$

$\Rightarrow p(\text{win} \mid 1,1) = 1$ because λ_{11} is the only card that loses when $(x, y) = (1,1)$;

$(x\ y)$	$p(\lambda_{00}\mid xy)$	$p(\lambda_{01}\mid xy)$	$p(\lambda_{10}\mid xy)$	$p(\lambda_{11}\mid xy)$
0 0	0	1/3	1/3	1/3
0 1	1/3	0	1/3	1/3
1 0	1/3	1/3	0	1/3
1 1	1/3	1/3	1/3	0

Here,

$$p(\lambda_{ij}) = \frac{1}{4}\left(\tfrac{1}{3}+\tfrac{1}{3}+\tfrac{1}{3}+0\right) = \tfrac{1}{4}$$

Although on average $p(\lambda_{ij})$ is $1/4$ similar to uniform, the total variation between $p(\lambda|xy)$ & $p(\lambda)$ is not zero.

$$d_{TV} = \frac{1}{2}\sum_{\lambda}\left|p(\lambda\mid 1,1) - p(\lambda)\right|$$

$$= \frac{1}{2}\Big(\left|p(\lambda_{00}\mid 1,1) - p(\lambda_{00})\right| + \left|p(\lambda_{01}\mid 1,1) - p(\lambda_{01})\right| + \left|p(\lambda_{10}\mid 1,1) - p(\lambda_{10})\right| + \left|p(\lambda_{11}\mid 1,1) - p(\lambda_{11})\right|\Big)$$

$$= \frac{1}{2}\left(\left|\tfrac{1}{3}-\tfrac{1}{4}\right| + \left|\tfrac{1}{3}-\tfrac{1}{4}\right| + \left|\tfrac{1}{3}-\tfrac{1}{4}\right| + \left|0-\tfrac{1}{4}\right|\right)$$

$$= \frac{1}{2}\left(3\left|\tfrac{1}{3}-\tfrac{1}{4}\right| + \left|0-\tfrac{1}{4}\right|\right) = \tfrac{1}{2}\left(\tfrac{1}{4}+\tfrac{1}{4}\right) = \tfrac{1}{4}$$

This d_{TV} is what we call as η_{bm}; in other words, if $\eta_{bm} = 1/4 \Rightarrow$ we can win the game using completely deterministic cards where the probability is dependent on (x, y).

Let $z = (x, y)$; the distribution of $z \Rightarrow q(z) = 1/4$ (prob.)

Let $f_z(\lambda)$ define win/loss (1/0) for a given card λ on z.

Total score $S[p] = \sum_z q(z) \underbrace{\sum_\lambda p(\lambda|z) f_z(\lambda)}_{\text{Avg win/loss as per } \lambda}$

Avg over different z.

Let $S_{ref} = \sum_z q(z) \sum_\lambda p(\lambda) f_z(\lambda)$

$\Delta = \sum_z q(z) \sum_\lambda \left(p(\lambda|z) - p(\lambda)\right) f_z(\lambda)$

$\Rightarrow S[p] = S_{ref} + \Delta$

$\sum_\lambda (p(\lambda|z) - p(\lambda)) f_z(\lambda) \leq \underbrace{\frac{1}{2} \sum_\lambda |p(\lambda|z) - p(\lambda)|}_{d_{TV}}$

$0 \leq f_z \leq 1$

$\therefore \Delta \leq \sum_z q(z) d_{TV}(p(\lambda|z), p(\lambda))$

$\therefore S[p] = S_{ref} + \Delta \leq S_{ref} + \underbrace{\sum_z q(z) d_{TV}(p(\lambda|z), p(\lambda))}_{\bar{\eta}(p)}$

$\Rightarrow S[p] \leq S_0 + \bar{\eta}(p) \leq S_0 + \eta_{bm}(p)$

$\eta_{bm} = \max_z d_{TV}(p(\lambda|z), p(\lambda))$

z: benchmark-instance or an input

$q(z)$: probability of z

λ : "hidden" deterministic strategy i.e., instruction cored

$\pi(\lambda)$: prob of choosing a specific card in benchmark-independent strategy

$f_z(\lambda) = \begin{cases} 1 & \text{win} \\ 0 & \text{loss.} \end{cases}$

For any z, the probability of losing under $\pi(\lambda)$ is

$$\ell_\pi(z) = \sum_\lambda \pi(\lambda)\left(1 - f_z(\lambda)\right)$$

Total score $S_0 = \sum_\lambda \pi(\lambda) f_z(\lambda) \Rightarrow \ell_\pi(z) = 1 - S_0$

For a given z, we want to move probability mass η from the cards that lose on z to those that win on z

$\frac{\pi(\lambda)}{S_0}$ → fraction of winning mass

So, we can construct $p(\lambda|z) = \underbrace{\pi(\lambda)}_{\text{uniform}} + \underbrace{\eta \frac{\pi(\lambda)}{S_0}}_{\text{move from losing to winning}}$

Similarly for a losing card we want

$$p(\lambda|z) = \pi(\lambda) - \eta \frac{\pi(\lambda)}{1 - S_0}$$

Combining

$$p(\lambda|z) = \pi(\lambda)\left[1 + \eta\left(\frac{f_z(\lambda)}{S_0} - \frac{1 - f_z(\lambda)}{1 - S_0}\right)\right]$$

If the card wins $f_z(\lambda) = 1 \Rightarrow (1 - f_z)$ term does not contribute.

If the card loses $f_z(\lambda) = 0 \Rightarrow f_z$ term does not contribute.

$$\sum_z q_z\, p(\lambda|z) = \pi(\lambda)\left[1 + \eta\left(\frac{\sum_z q_z f_z(\lambda)}{S_0} - \frac{\sum_z q_z[1-f_z(\lambda)]}{1-S_0}\right)\right] \quad \textcircled{*}$$

q is fn of z; f_z is fn of λ

$$\sum_z q_z = 1 \qquad S_0 = \sum_z q_z \sum_\lambda \pi(\lambda) f_z(\lambda)$$

$\sum_z q(z) f_z(\lambda)$ is avg score of an individual strategy or card λ

$$\sum_z q(z) f_z(\lambda) = F(\lambda)$$

we must have $F(\lambda) \leq S_0$ or S_{ref} or $S_{classical}$

But by definition $S_0 = \sum_z q_z \sum_\lambda \pi(\lambda) f_z(\lambda)$

$$= \sum_\lambda \pi(\lambda) \underbrace{\sum_z q(z) f_z(\lambda)}_{F(\lambda)} = \sum_\lambda \pi(\lambda) F(\lambda)$$

$F(\lambda) \leq S_0$ & $\sum_\lambda \pi(\lambda) F(\lambda) = S_0 \} \Rightarrow F(\lambda) = S_0$

$\therefore \sum_z q(z) f_z(\lambda) = S_0 \Rightarrow \sum_z q(z)(1 - f_z(\lambda)) = 1 - S_0$

Substitute into $\textcircled{*}$

$$\hookrightarrow \sum_z q(z) p(\lambda|z) = \pi(\lambda)\left\{1 + \eta\left[\frac{S_0}{S_0} \frac{1-S_0}{1-S_0}\right]\right\} \quad (\text{the bracket} \to 0)$$

$$= \pi(\lambda).$$

If we distribute classical strategies so that they make mistakes equally often (e.g., CHSH game strategies losing on at least one (x, y)), then even a small correlation between the hidden resource & the input can be used with maximum efficiency to increase the classical score.

For example, see the table on 1-2; Any card / strategy loses on one (x, y). So, the winning probability is $3/4$ irrespective of which input is selected.

This is what "loss equalization" means physically.

In the ideal scenario, the card selection is independent of input But, suppose the hidden resource is slightly correlated with the settings i.e., the probability of selecting a card can vary depending on the input.

Suppose the input is $(x, y) = (0, 0)$

In the table on page 1-2, the card λ_{00} loses on $(0,0)$. The uniform probability of A & B drawing λ_{00} is $1/4$ in Table on 1-2.

Now, allow some dependence of card probability on the input; we do this through η; let's say $\eta = 1/10$

Based on the input & η, we can redistribute the probabilities. For example, when the input is $(0, 0)$ reduce the probability of λ_{00} from $1/4$ to $1/4 - \eta$ $= 1/4 - 1/10 = 0.15$; At the same time, we increase the probabity of 3 other cards $\lambda_{01}, \lambda_{10}$, & λ_{11} by the same amount (collectively); so each card $\lambda_{01}, \lambda_{10}, \lambda_{11}$ increases from $0.25 \rightarrow 0.28$.

Card	a	b	$a \oplus b$	xy	losing case	$p(\lambda)$	$p(\lambda \mid x,y)$ / ex: $p(\lambda \mid 0,0)$
λ_{00}	$1 \oplus x$	y	1	0	(0,0)	1/4	$1/4 - \eta$
λ_{01}	0	y	1	0	(0,1)	1/4	$1/4 + \eta/3$
λ_{10}	x	0	1	0	(1,0)	1/4	$1/4 + \eta/3$
λ_{11}	0	0	0	1	(1,1)	1/4	$1/4 + \eta/3$.

$$s[p] = \sum_z q(z) \sum_\lambda p(\lambda \mid z) f_z(\lambda)$$

$q(z) = 1/4$ for $(x,y) = (0,0)\ (0,1)\ (1,0)$ & $(1,1)$.

Case $p(\lambda \mid z) = 1/4 = p(\lambda)$ irrespective of z

$$s[p] = \underbrace{1/4}_{\text{probability that } z=(0,0)} \Big(\underbrace{1/4}_{\text{prob. of losing card } \lambda_{00}} \cdot 0 + \underbrace{3/4}_{\text{prob. of picking winning cards on } (0,0) \{\lambda_{01}, \lambda_{10}, \lambda_{11}\}} \cdot 1 \Big) + \underbrace{1/4}_{\text{prob of } (0,1)} \Big(\underbrace{1/4}_{\text{prob of } \lambda_{01}} \cdot 0 + \underbrace{3/4}_{\text{prob of } \lambda_{00}, \lambda_{10}, \lambda_{11}} \cdot 1 \Big) + \cdots$$

two more such terms for 10 & 11

$$= 4 \cdot 1/4 \left(1/4 \cdot 0 + 3/4 \right) = \underline{3/4}$$

Case $p(\lambda \mid z) \neq p(\lambda)$

$$s[p] = \underbrace{1/4}_{\text{prob of } z=(0,0)} \Big(\underbrace{(1/4 - \eta)}_{\text{prob of } \lambda_{00}} 0 + (3/4 + \eta) \cdot 1 \Big) + \text{three more such terms}$$

$$= 4 \cdot 1/4 \left(3/4 + \eta \right)$$

↳ we adjusted based on the input

$$= \underline{3/4 + \eta}$$

Card	$p(\lambda \mid 0,0)$	$p(\lambda \mid 0,1)$	$p(\lambda \mid 1,0)$	$p(\lambda \mid 1,1)$	Avg
λ_{00}	$1/4 - \eta$	$1/4 + \eta/3$	$1/4 + \eta/3$	$1/4 + \eta/3$	$1/4$
λ_{01}	$1/4 + \eta/3$	$1/4 - \eta$	$1/4 + \eta/3$	$1/4 + \eta/3$	$1/4$
λ_{10}	$1/4 + \eta/3$	$1/4 + \eta/3$	$1/4 - \eta$	$1/4 + \eta/3$	$1/4$
λ_{11}	$1/4 + \eta/3$	$1/4 + \eta/3$	$1/4 + \eta/3$	$1/4 - \eta$	$1/4$

If we average the Card distributions over all four inputs, the average is still $1/4$; so, overall distribution of hidden resource remains unchanged although the resource is correlated with particular input.

→ This is the "marginal-consistency" requirement.

define $\eta_{bm} = \max_z d_{TV}(p(\lambda \mid z), p(\lambda))$

The maximum score we can achieve with η is called the "frontier"

$$S_\eta = \sup_{\eta_{bm} \le \eta} S[p] = \min\{1, s_0 + \eta\} \quad \text{for every } \eta \ge 0$$

Theorem 1: part (a) $S_\eta = \min\{1, s_0 + \eta\}$ for $\eta > 0$

form an interval $[0, \eta_*]$ with $\eta_* \le \max_z \min_{\pi \text{ optimal}} \ell_\pi(z)$.

where $\ell_\pi(z) = \sum_z \underbrace{\pi(\lambda)}_{\text{unbiased prob.}} \left(1 - f_z(\lambda)\right)$ → Loss probability.

The input with least losing mass limits how much probability can be transferred from losses to wins on every input simultaneously

Theorem 2 answers the case where the inputs occur with unequal probabilities. What is the exact score attainable with a dependence of η?

Single-loss game:
1) Every admissible deterministic card/strategy loses on at least one input
2) For each input i, there $\exists$ S_i, a specific card/strategy that loses only on input i

Let $q_i = Pr(z = i)$ $\quad q_{min} = \min_i q_i$ $\quad q_{max} = \max_i q_i$

$$S = \sum_j q_j \, Pr(win|j)$$

$$S[S_i] = \sum_{i \neq j} q_j \underbrace{Pr(win|j)}_{1} + \sum q_i \overbrace{Pr(win|i)}^{\to 0}$$

↑ we choose S_i, which means $Pr(win|j) = 0$ when $q_j = q_i$

but $\sum_j q_j = 1 \Rightarrow \sum_{i \neq j} q_j = 1 - q_i$

$$\therefore S[S_i] = \sum_{i \neq j} q_j$$

$$\therefore S[S_i] = 1 - q_i$$

So, the best card/strategy will be that which belongs to q_{min} i.e., choose the card such that the input it loses on has the least probability

$$\therefore S_0 = \max_i S[S_i] = 1 - q_{min}$$

The bias in $q(z)$ raises the independent win probability because the unavoidable loss can be placed on a rarer input.

Define $p_i(j) = Pr($ strategy j / input $i)$ } the probability of picking a specific card or strategy

The "marginal" is ~~summed over or~~ averaged over

Card	(0,0)	(0,1)	(1,0)	(1,1)	Avg → Π_j
λ_{00}	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$ → This is unbiased
λ_{01}	$\frac{1}{4}-\eta$	$\frac{1}{4}+\frac{\eta}{3}$	$\frac{1}{4}+\frac{\eta}{3}$	$\frac{1}{4}+\frac{\eta}{3}$	$\frac{1}{4}$ → Here we ~~performed the swap~~ redistributed bench mark dependence. However, the avg must still be $\frac{1}{4}$ This is marginal consistency.
λ_{10}					
λ_{11}					

$$\Pi_j = \sum_i q_i p_i(j) \qquad i \to \text{summed over all inputs.}$$

In a single loss game, by definition (2) on 1-10, we have a loss on input i for strategy S_i.

$\therefore$ $Pr($ loss $/\ i)$ is same as the probability of picking strategy S_i

$$\therefore \quad Pr(\text{loss} / i) = p_i(i)$$

If $S[p]$ is the winning score, $1 - S[p]$ is the loss.

The average loss is $\sum_i q_i p_i(i)$

$$\therefore \quad 1 - S[p] = \sum_i q_i p_i(i).$$

$$\boxed{S[p] = 1 - \sum_i q_i p_i(i)}$$

To maximize $S[p]$ minimize $\sum_i q_i p_i(i)$; we already know the optimum from (#) on 1-10; $\boxed{S_0 = 1 - q_{min}}$

Compare the prob. $p_i(i)$ with its marginal π_i

recall $\pi_j = \sum_i q_i p_i(j)$

Define $u_i = \pi_i - p_i(i)$

our benchmark dependence is $d_{TV}(p_i(\cdot), \pi_\cdot) \le \eta$ (*)

$$\sum_j p_i(j) = 1 \quad ; \quad \sum_j \pi_j = 1$$

$$\sum_{j \neq i} (p_i(j) - \pi_j) = (1 - p_i(i)) - (1 - \pi_i) = u_i$$

$$d_{TV}(p_i, \pi) = \frac{1}{2}\left[\sum_j |p_i(j) - \pi_j|\right]$$

$$= \frac{1}{2}\left(\sum_{j \neq i} |p_i(j) - \pi_j| + |p_i(i) - \pi_i|\right)$$

$$\ge \frac{1}{2}(|u_i| + |u_i|) = |u_i|$$

$$\sum_{j \neq i} |p_i(j) - \pi_j| \ge \left|\sum_{j \neq i} (p_i(j) - \pi_j)\right| = |u_i|$$

$\therefore$ $\boxed{d_{TV} \ge |u_i|}$ ⟶ (#)

Combine *&# ⟶ $\boxed{|u_i| \le d_{TV}(p_i, \pi) \le \eta}$

$p_i(i) = \pi_i - u_i$ $\quad \left(\sum_i p_i(i) = \sum_i \pi_i - \sum_i u_i = 1 - \sum_i u_i\right)$

↳ sum of all losing probabilities (**)

Average loss: $\sum_i q_i p_i(i)$.

$$1 - S[p] = \sum_i q_i\, p_i(i)$$

Every $q_i \geq q_{min}$ of every $p_i(i) \geq 0$

$$\therefore \quad 1 - S[p] \geq q_{min} \underbrace{\sum_i p_i(i)}_{\text{total less. (see (**))}}$$

$$\boxed{1 - S[p] \geq q_{min}\left(1 - \sum_i u_i\right)} \longrightarrow (\#)$$

we know $u_i \leq \eta$

Suppose we have m strategies $S_1, S_2, S_3, \ldots S_m$.

we can define $u_1, u_2, \ldots u_m$, where

$u_i = \underbrace{\pi_i}_{\text{marginal}} - \Pr(\underbrace{\text{strategy } i}_{\text{losing strategy}} \text{ is picked on input } i)$

$\Rightarrow \sum_{i=1}^{m} q_i \leq \sum_{i=1}^{m} \eta = m\eta$

now $\sum_i u_i \leq m\eta$

Consider "h" as the input with highest probability

$$Q = q_h = q_{max}$$

$$\pi_h = \sum_k q_k\, p_k(h) = q_h\, p_h(h) + \sum_{k \neq h} q_h\, p_k(h)$$

$$\boxed{\pi_h = Q\, p_h(h) + \sum_{k \neq h} q_k\, p_k(h)} \longrightarrow (!!)$$

$\sum_k q_k = 1 \Rightarrow \underbrace{q_h}_{Q} + \sum_{h \neq k} q_k = 1 \Rightarrow Q + \sum_{h \neq k} q_k = 1$

$$\hookrightarrow \boxed{Q\,\pi_h + \sum_{h \neq k} q_k\, \pi_h = \pi_h} \longrightarrow (*)$$

(1.1) - (*)

↳

$$Q(\underbrace{p_h(h) - \pi_h}_{-u_h}) + \sum_{h \neq k} q_k (p_k(h) - \pi_h) = 0$$

$$\therefore \underbrace{\sum_{h \neq k} q_k (p_k(h) - \pi_h)}_{\text{net increase of the same strategy on other inputs}} = \underbrace{Q u_h}_{\text{weighted reduction of strategy } s_h \text{ on it's losing input}}$$

$$|p_i(j) - \pi_j| \leq d_{TV}(p_i, \pi) \leq \eta$$

$$\therefore \left| \sum_{h \neq k} q_k (p_k(h) - \pi_h) \right| \leq \sum_{h \neq k} q_k |p_k(h) - \pi_h| \leq \sum_{h \neq k} q_k \eta$$

$$\therefore Q|u_h| \leq \sum_{k \neq h} q_k \eta = \eta \sum_{k \neq h} q_k = \eta(1-Q)$$

$$\therefore |u_h| \leq \eta \left(\frac{1-Q}{Q} \right)$$

Thus $u_h \leq \min \left\{ \eta, \eta \left(\frac{1-Q}{Q} \right) \right\}$.

If $q \le 1/2 \Rightarrow \frac{1-q}{q} \ge 1 \Rightarrow u_h \le \min\left\{\eta, \eta\left(\frac{1-q}{q}\right)\right\}$ can not be tightened further as $\eta < \eta\left(\frac{1-q}{q}\right)$

If $q > 1/2 \quad \frac{1-q}{q} < 1 \Rightarrow u_h \le \min\left\{\eta, \eta\left(\frac{1-q}{q}\right)\right\}$ can be lower than η

$$\therefore \sum_i u_i = \underbrace{\sum_{i \ne h} u_i}_{\le \sum_{i=1}^{m-1} (m-1)\eta} + u_h \le \min\left\{\eta, \eta\left(\frac{1-q}{q}\right)\right\}$$

$$\therefore \sum_i u_i \le \underbrace{\left(m - 1 + \min\left\{1, \frac{1-q}{q}\right\}\right)}_{m_{eff}} \eta$$

$$\Rightarrow m_{eff} = \begin{cases} m & q \le 1/2 \\ m - 2 + \frac{1}{q} & q > 1/2 \end{cases}$$ effective input number

$$\sum_i u_i \le m_{eff}\,\eta$$

Go back to (#) on 1-13

$$1 - S[p] \ge q_{min}\left(1 - \sum_i u_i\right) \ge q_{min}(1 - m_{eff}\,\eta)$$

$$S[p] \le \underbrace{1 - q_{min}}_{S_0} + m_{eff}\, q_{min}\,\eta$$

$$\therefore S_\eta \le \min\{1, S_0 + m_{eff}\, q_{min}\}.$$

we want to construct a policy that maximizes S_η

choose an input $k \in \arg\min_i \{q_i\}$ choose any one of the indices from the set of indices where q is minimum.

$S[p] \leq 1 - q_{min} + m_{eff} q_{min} \eta$ } from 1-15

This is maximized to 1 when $\eta = \frac{1}{m_{eff}}$

so, we want to construct a policy such that $0 \leq \eta \leq 1/m_{eff}$

Let us label "u_i" for this specific policy that maximizes $S[p]$ as d_i

If $q_{max} \leq 1/2$, choose $d_i = \eta$ for every input

If $q_{max} > 1/2$, choose $d_i = \eta$ for all inputs except $i \neq h$ h being the index where q_{max} occurs.

$$d_h = \frac{1 - q_{max}}{q_{max}} \eta = \frac{1-\alpha}{\alpha} \eta$$

Now, Sum d_i over all strategies. (m in total)

$$\sum_{i=1}^{m} d_i = m_{eff} \eta$$

Recall

$$m_{eff} = \begin{cases} m & \text{for } q_{max} \leq 1/2 \\ m - 2 + \frac{1}{q_{max}} & \text{for } q_{max} > 1/2 \end{cases}$$

Define residual $R = 1 - m_{eff} \eta$

choose the marginal $\pi_i = d_i + R \underbrace{1_{\{i=k\}}}_{\text{like a dirac delta prob.}}$

$$\sum_i \pi_i = \sum_i d_i + \sum_i R 1_{\{i=k\}} = \sum_i d_i + R$$

$$= m_{eff} \eta + 1 - m_{eff} \eta = 1$$

So $\eta = \frac{1}{m_{eff}}$ maximizes the score while obeying marginal consistency.

For a target score $S > S_0$, we need $\eta_{req}(S) = \frac{S - S_0}{m_{eff}\, q_{min}}$;

q_{min} → determines how costly the remaining loss is

q_{max} → determines whether marginal consistency restricts how much loss can be removed.

What happens if the experimental settings are sampled according to one distributions, but the reported Bell score weights those settings according to a different distribution?

Suppose we have m inputs $z \in \{1, 2, \ldots m\}$

we used $q(z)$ → looking at how frequent a given input is
→ how strongly the performance on z → contributes to the final score

~~In CHSH~~, $m=4$; $z \in \{00, 01, 10, 11\}$

$q(00), q(01), q(10), \& q(11)$ → Sampling probabilities.

Final Score: $S = \frac{1}{4}\{P(\text{win}|00) + P(\text{win}|01) + P(\text{win}|10) + P(\text{win}|11)\}$

$\frac{1}{4}$ → we are assigning equal weight; this in general need not be true.

Let us say we weigh the scores by $w(z)$;

If we want to report the actual-game winning fraction $\omega = q$

If we instead report the conventional $1/4$ avg then $w \neq q$

we are sampling the inputs with $q(z)$ but giving equal weight $1/4$ when we use the traditional expression for S.

Thm 2 we saw $S_{\eta,q} = \min\{1, 1 - q_{min} + m_{eff}(q)\, q_{min}\, \eta\}$.

Now, if we want to use $q(z)$ for sampling but wish to report the conventional score with $\omega(z) = 1/4$

then $S_{\eta,\omega} = \min\{1, 1 - 1/4 + \frac{m_{eff}(q)}{4}\eta\}$

$= \min\{1, 3/4 + \frac{m_{eff}(q)}{4}\eta\}$.

We have used own marginal consistency so far. This means

$$\sum_z q(z)\, P(\lambda|z) = \pi(\lambda)$$

For example, consider the following table

Card	$P(\lambda\|00)$	$P(\lambda\|01)$	$P(\lambda\|10)$	$P(\lambda\|11)$
λ_{00}	$1/4 - \eta$	$1/4 + \eta/3$	$1/4 + \eta/3$	$1/4 + \eta/3$
λ_{01}	$1/4 + \eta/3$	$1/4 - \eta$	$1/4 + \eta/3$	$1/4 + \eta/3$
λ_{10}	$1/4 + \eta/3$	$1/4 + \eta/3$	$1/4 - \eta$	$1/4 + \eta/3$
λ_{11}	$1/4 + \eta/3$	$1/4 + \eta/3$	$1/4 + \eta/3$	$1/4 - \eta$

In this table, all the columns sum to 1

i.e., $\sum_\lambda P(\lambda|z) = 1$

This always needs to hold true.

But, look at how we distributed η

We distributed η equally such that if you look at a row, you get the avg to be $1/4$; This condition "own marginal consistency" is imposed; we need not have this. We can choose that each row averages to some other value (i.e., $\sum_z q(z) P(\lambda|z) \neq \pi(\lambda)$) while having $\sum_\lambda P(\lambda|z) = 1$

In other words, we can distribute η unequally.

Card	$P(\lambda\|00)$	$P(\lambda\|01)$	$P(\lambda\|10)$	$P(\lambda\|11)$
λ_{00}	$1/4-\eta$	$1/4+\eta/3$	$1/4+\eta/2$	$1/4$
λ_{01}	$1/4$	$1/4-\eta$	$1/4+\eta/2$	$1/4+\eta/2$
λ_{10}	$1/4+\eta/2$	$1/4+\eta/3$	$1/4-\eta$	$1/4+\eta/2$
λ_{11}	$1/4+\eta/2$	$1/4+\eta/3$	$1/4$	$1/4-\eta$

Notice that in this new way of distributing η, we do not satisfy $\sum_z q(z) P(\lambda|z) = \pi(\lambda) = 1/4$ any more although each column still sums to 1

What changes if we allow the reference distribution of hidden variables to be chosen "freely", rather than requiring it to equal the own marginal distribution?

Let's call the free reference as $G(\lambda)$

winning probability on an input z: $f_G(z) = \sum_\lambda G(\lambda) f_z(\lambda)$

losing prob: $\ell_G(z) = 1 - f_G(z)$.

If we allow η benchmark dependence, $d_{TV}(p_z, G) \le \eta$

$$\therefore \sum_\lambda p_z(\lambda) f_z(\lambda) \le \underbrace{\sum_\lambda G(\lambda) f_z(\lambda)}_{\longrightarrow f_G(z)} + \eta$$

$$\therefore P(\text{win}|z) \le \min\{ f_G(z) + \eta, 1 \}.$$

$$\Rightarrow S[p] \le \sum_z q(z) \min\{ f_G(z) + \eta, 1 \}$$

Optimize over all $G(\lambda)$ $\rightarrow$ $S_\eta^{\text{free}} \le \max_{G \in \Delta(O)} \sum_z q(z) \min\{ f_G(z) + \eta, 1 \}$

$\Delta(O)$ denotes all probability distributions over the admissible deterministic strategies.

Comparison with Hall, MDL: Our measure compares each conditional distribution with the unconditional marginal

$$\eta_{bm} = \max_z d_{TV}(P_z, \pi)$$

Hall instead compares conditional distributions directly

$$\mathcal{M} = 2 \max_{z,z'} d_{TV}(P_z, P_{z'})$$

So, the physical questions are different.

Our measure answers: How far can the resource distribution shift (when I condition on a specific input) relative to its overall distribution.

Hall is answering a different question. How different can the resource distributions be between two possible inputs?

Our measure η_{bm}: worst-case distance to the actual marginal

Hall's $\mathcal{M}$: largest pairwise difference between conditional laws

MDL ℓ: minimum permitted setting likelihood ratio
small $\ell \to$ more dependence.
At $\ell = 1 \to$ no dependence.

$I(Z;\Lambda)$: mutual information carried about the input.

They all quantify departures from "independence" but they impose different restrictions on the allowed classical models. One of them is not universally better than the other. It depends upon the specific physical situation of statistical constraint which one we wish to use.

Suppose A & B perform a CHSH experiment; But, one setting pair is selected much more often than the other three. A classical HV model wants to exploit the information about settings to improve the score. Our measure compares HV distribution for each setting with the overall distribution of HV

Suppose one setting is used in more than half the experiments. If the classical model changes its HV distribution for this setting, to increase $Pr(win|z)$, this change has a large influence on the overall distribution; As other settings occur less frequently, they cannot fully compensate for the change with in the same dependence allowance. In MDL, every measurement setting retains some minimum probability. Our measure has an additional restriction arising from "own marginal" consistency. So, it captures "Setting-bias" effect that MDL does not

2→1 RAC:

A receives two independent, uniformly random bits.

$x = (x_0, x_1) \in \{0,1\}^2$

B receives a question $y \in \{0,1\}$

B's goal is to learn A's bit specified by y

i.e., if $y=0$ → B should output x_0

$y=1$ → B should output x_1

A can send B only one bit. After receiving the bit "a" from A, B outputs "b" based on y & a; if $b = x_y$ it is a win.

Best classical strategy: A always sends her first bit x_0 to B. B outputs the received bit.

for $y=0$

x_0	x_1	a	y	b	result
0	0	0	0	0	win
0	1	0	0	0	win
1	0	1	0	1	win
1	1	1	0	1	win
0	0	0	1	0	lose
0	1	0	1	0	lose
1	0	1	1	1	win
1	1	1	1	1	win

$P(\text{win} \mid y=0) = 1 \qquad P(\text{win} \mid y=1) = 1/2$

Assuming that $P(y=0) = P(y=1) = 1/2$

$$S_0 = \frac{1}{2} P(\text{win} \mid y=0) + \frac{1}{2} P(\text{win} \mid y=1)$$

$$= \frac{1}{2}(1) + \frac{1}{2}\left(\frac{1}{2}\right) = \frac{3}{4}.$$

Quantum Strategy : Now A can share 1 qubit instead of 1 classical bit; Suppose A prepares a state based on x_0, x_1; the Bloch sphere vector $\vec{r}_x$ for this can be

$$\vec{r}_x = \frac{1}{\sqrt{2}}\left((-1)^{x_0}\hat{z} + (-1)^{x_1}\hat{x}\right)$$

B measures along Z if $y=0$, along X if $y=1$

If measurement is $+1 \rightarrow b=0$

$-1 \rightarrow b=1$

For a qubit with Bloch vector $\vec{r}$

$$\rho = \frac{1}{2}\left(\mathbb{I} + \vec{r}\cdot\vec{\sigma}\right)$$

A projective measurement along a unit vector $\hat{n}$ is

$$\Pi_s = \frac{1}{2}\left(\mathbb{I} + s\,\hat{n}\cdot\vec{\sigma}\right) \qquad s \in \{+1, -1\}$$

$$\therefore \quad P(s) = \mathrm{Tr}(\rho\Pi_s) = \frac{1}{4}\mathrm{Tr}\left(\left(\mathbb{I} + \vec{r}\cdot\vec{\sigma}\right)\left(\mathbb{I} + s\,\hat{n}\cdot\vec{\sigma}\right)\right)$$

$$= \frac{1}{4}\mathrm{Tr}\Big(\underbrace{\mathbb{I}}_{2} + \underbrace{s\,\hat{n}\cdot\vec{\sigma} + \vec{r}\cdot\vec{\sigma}}_{\mathrm{Tr}(\sigma_i)=0} + s(\vec{r}\cdot\vec{\sigma})(\hat{n}\cdot\vec{\sigma})\Big) \qquad \mathrm{Tr}(\sigma_i\sigma_j) = 2\delta_{ij}$$

$$= \frac{1}{4}\left(2 + 2s\,\vec{r}\cdot\hat{n}\right) = \frac{1}{2}\left(1 + s\,\vec{r}\cdot\hat{n}\right)$$

$\vec{r}_x \cdot \hat{n}$ ⟶ $y=0 \Rightarrow \vec{r}_x\cdot\hat{z} = \dfrac{(-1)^{x_0}}{\sqrt{2}}$

↘ $y=1 \Rightarrow \vec{r}_x\cdot\hat{x} = \dfrac{(-1)^{x_1}}{\sqrt{2}}$

$$P(s\,|\,y=0) = \frac{1}{2}\left(1 + s\left(\frac{-1}{\sqrt{2}}\right)^{x_0}\right) \qquad P(s\,|\,y=1) = \frac{1}{2}\left(1 + s\left(-\frac{1}{\sqrt{2}}\right)^{x_1}\right)$$

when $y=0$, B wins when $S = (-1)^{x_0}$

$y=1$, B win when $S = (-1)^{x_1}$

$$\therefore P(\text{win}\mid y=0) = \frac{1}{2}\left(1 + (-1)^{x_0}\frac{(-1)^{x_0}}{\sqrt{2}}\right) = \frac{1}{2}\left(1+\frac{1}{\sqrt{2}}\right)$$

$$P(\text{win}\mid y=1) = \frac{1}{2}\left(1 + (-1)^{x_1}\frac{(-1)^{x_1}}{\sqrt{2}}\right) = \frac{1}{2}\left(1+\frac{1}{\sqrt{2}}\right)$$

$$\therefore S_Q = \frac{1}{2}\left(1+\frac{1}{\sqrt{2}}\right) > S_0$$

Could classical one-bit reproduce $S_{Quantum}$ if its shared hidden resource were correlated with the inputs?

Model 1: $P(\lambda\mid x,y)$ HV dependent on both x & y

This is similar to CHSH game

$$S_\eta = \min\left\{1, \frac{3}{4}+\eta\right\}$$

$$\therefore \eta_{req} = S_Q - S_0 \cong 0.1$$

Model 2: $P(\lambda\mid y)$ dependent only on Bob's input.

A receives (x_0, x_1) & a shared hint y'. A sends $a = x_{y'}$; B outputs $b = a = x_{y'}$

if $y'=y$ → win

$$S = P(y'=y)\times 1 + P(y'\neq y)\times \frac{1}{2}$$

if $P(y'=y) = \frac{1}{2}$ ⇒ we get $S = S_0 = \frac{3}{4}$

if $P(y'=y) = \frac{1}{2}+\eta$ ⇒ $P(y'\neq y) = \frac{1}{2}-\eta$

$\Rightarrow S = \frac{3}{4} + \eta/2$

Same for $P(\lambda\mid x)$ also.